AI (ARTIFICIAL INTELLIGENCE

CO1 - Assessment tool - 3

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Slot-B

1. Intelligent Agents in a smart home System :-

An intelligent agent senses its environment (motion, temperature, time) and acts ont it (light, HVAC, locks)

=> Simple reflex: acts on the cwoent input using if-then rules, with no memory. Example: motion at night -> light on; forced door -> alarm

=> Model-based: keeps an interval state of the world. Example · remembers who is home and the last thermostat setting even when sensors quiet

=> Goal-based: plans actions to reach a goal.

=> Utility-based: picks the action with the best trade-off between comfort, energy cost and safety

=> Best choice: a hybrid agent Reflex rules give instant security responses, the model tracks occupancy - goals set targets and utility with learning balance comfort and energy saving.

2. Uninformed Search for Maze Robot:

- Uninformed search uses no knowledge of where the exit is, so it suits an unknown maze (Cells = states, moves = actions),

=> UCS: expands the lowest path-cost node first.

Complete and optimal, so it finds the cheapest route

Drawback explores widely and needs exponentials time and memory

=> IDS: repeats depth-limited DFS with increasing limits

Complete, optimal for equal step costs, but space only O(bd)

Drawback: repeats upper levels and ignores unequal cost

=> Agent design: a simple reflex robot can loop forever.

A model based robot remembers visited cells and build a map.

=> Best combination: a model based, goal-based agent using IDS to explore (low memory, finds the exit) than a utility based agent using UCS.